Modern Hopfield Networks with Latent-Space Retrieval

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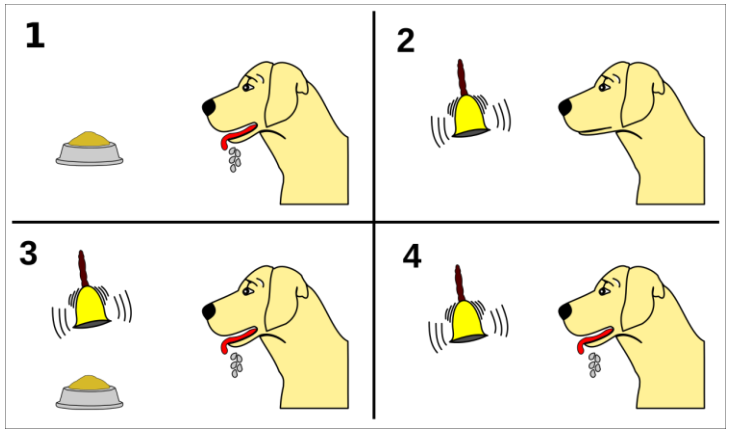
Hopfield Network

> Associative memory

Hopfield network

Modern Hopfield Networks



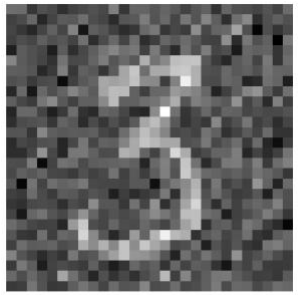
Hopfield Network

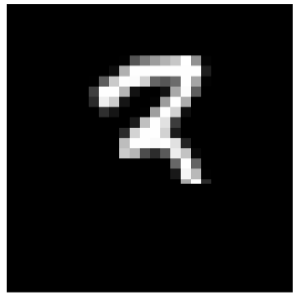
> Associative memory

Hopfield network

Modern Hopfield Networks

Train memory





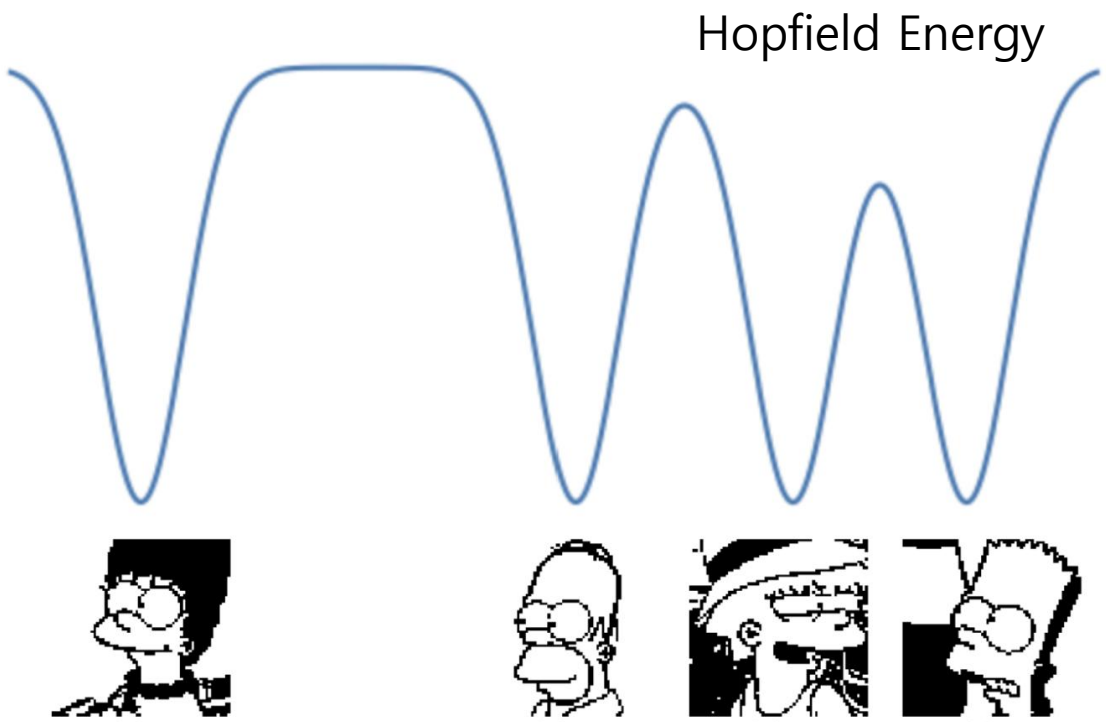
Recalled memory

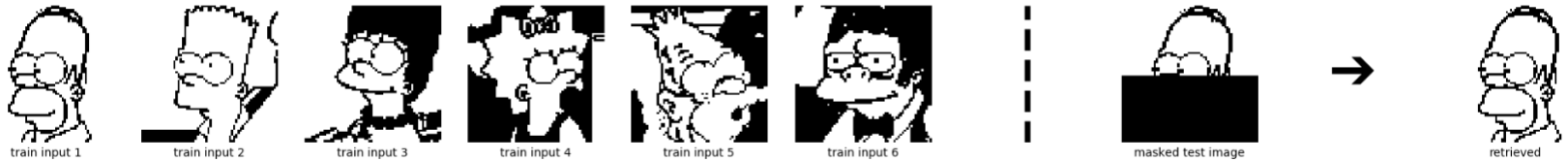
Hopfield Network

> Associative memory

Hopfield network

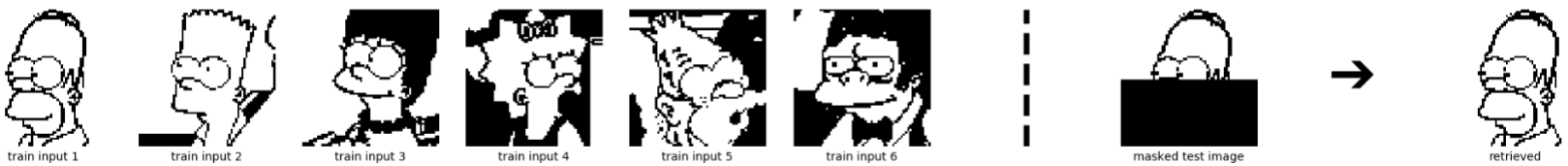
Modern Hopfield Networks





Sacco, et al., "Requirements for self organization", zenodo, 2023
Ramsauer et al., Hopfield Networks is All You Need. 2020

Hopfield Network



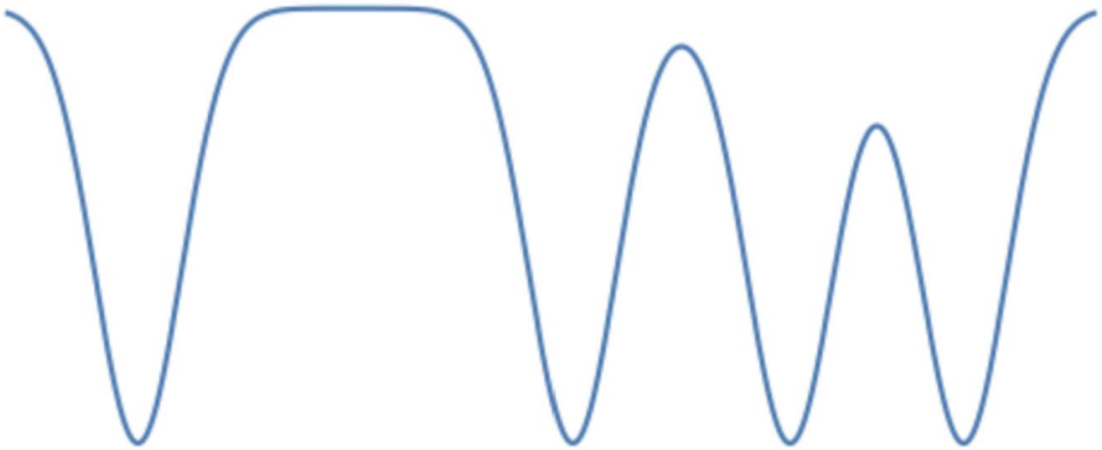
Associative memory

> Hopfield network

Modern Hopfield Networks

$$\{\boldsymbol{x}_i\}_{i=1}^N$$

$$\boldsymbol{x}_i \in \{-1, 1\}^d,$$



Hopfield Energy

$$\mathrm{E} = -\frac{1}{2}\boldsymbol{\xi}^T \boldsymbol{W} \boldsymbol{\xi} + \boldsymbol{\xi}^T \boldsymbol{b}$$

$$\boldsymbol{W} = \sum_i^N \boldsymbol{x}_i \boldsymbol{x}_i^T \ .$$

$$\boldsymbol{\xi}^{t+1} = \mathrm{sgn}(\boldsymbol{W}\boldsymbol{\xi}^t - \boldsymbol{b}) \ ,$$

Hopfield Network

Associative memory

> Hopfield network

Modern Hopfield Networks

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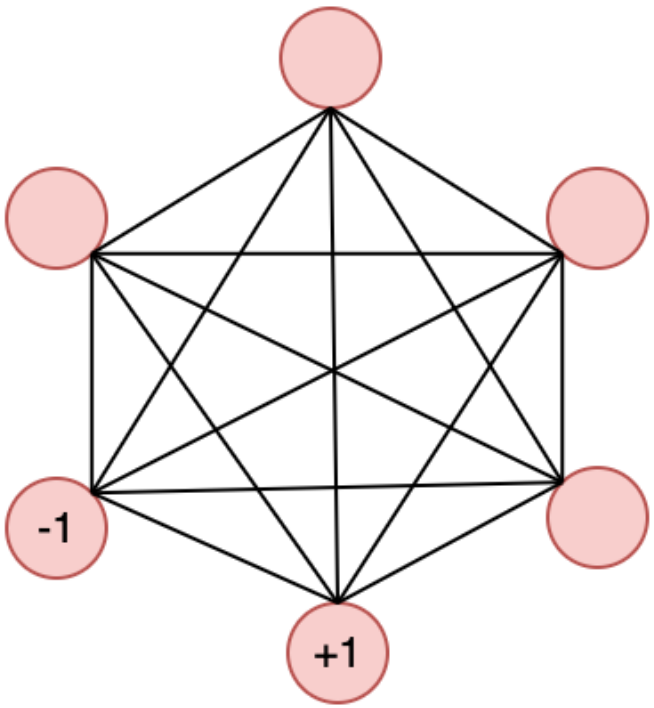


$$\mathrm{E} = -\frac{1}{2}\boldsymbol{\xi}^T \boldsymbol{W} \boldsymbol{\xi} + \boldsymbol{\xi}^T \boldsymbol{b}$$

Hebbian Learning ! $$\boldsymbol{W} = \sum_i^N \boldsymbol{x}_i \boldsymbol{x}_i^T \ .$$

$$\boldsymbol{\xi}^{t+1} = \mathrm{sgn}(\boldsymbol{W}\boldsymbol{\xi}^t - \boldsymbol{b}) \ ,$$

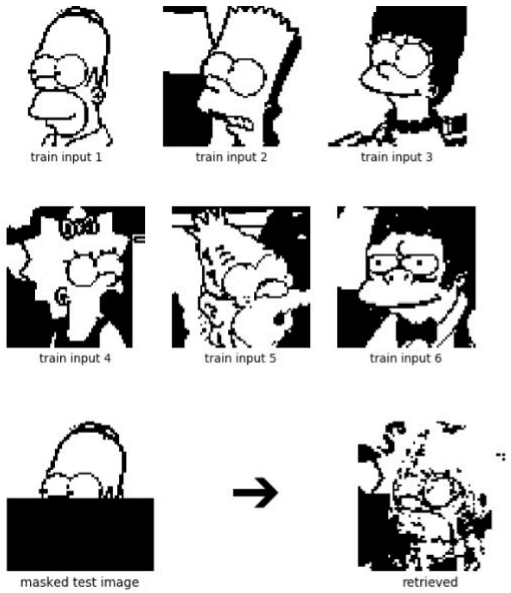
Changsin Lee., "Understanding the Hopfield Network with Examples", medium, 2024

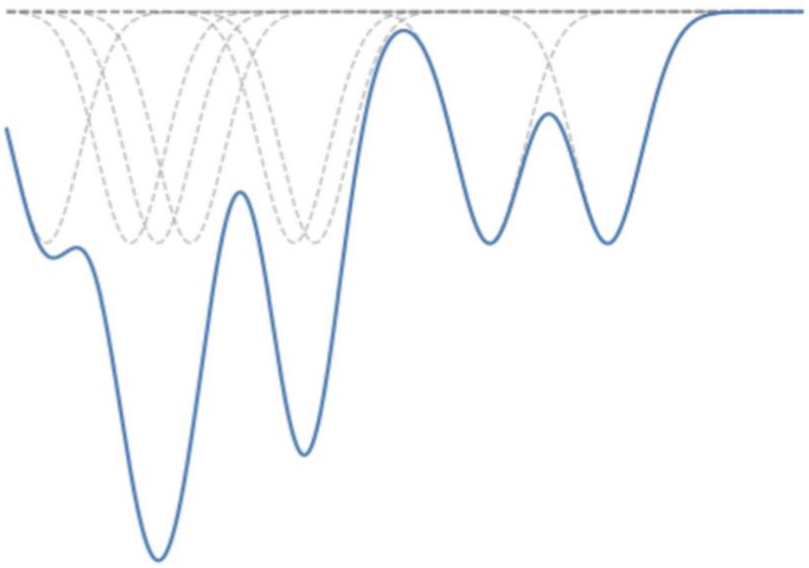
Hopfield Network

Associative memory

> Hopfield network

Modern Hopfield Networks





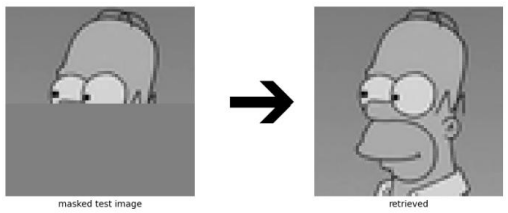
$$C \cong \frac{d}{2\log(d)}\,, \qquad C \cong 0.14d\,.$$

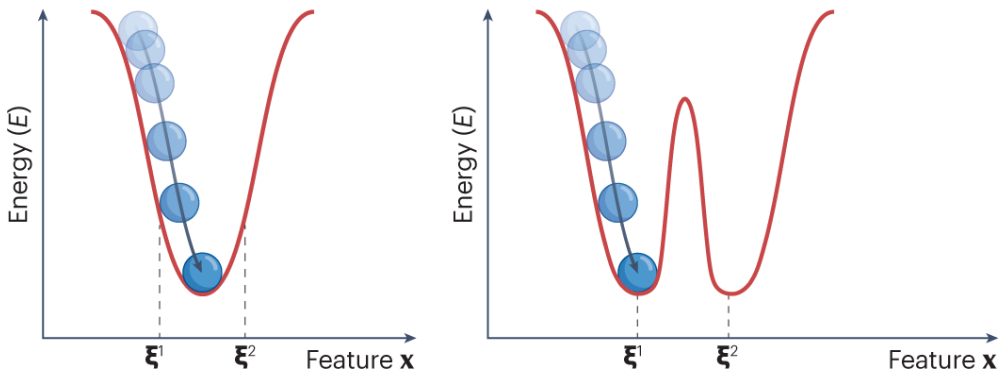
Hopfield Network

Associative memory

Hopfield network

> Modern Hopfield Networks





$$\mathrm{E} = -\frac{1}{2}\boldsymbol{\xi}^T \boldsymbol{W}\boldsymbol{\xi} + \boldsymbol{\xi}^T \boldsymbol{b}$$

$$\mathrm{E} = -\sum_{i=1}^{N} \exp(\boldsymbol{x}_i^T \boldsymbol{\xi}) \ ,$$

$$\boldsymbol{\xi}^{t+1} = \mathrm{sgn}(\boldsymbol{W}\boldsymbol{\xi}^t - \boldsymbol{b}) \ ,$$

$$\boldsymbol{\xi}^{t+1} = \boldsymbol{X}\mathrm{softmax}(\beta \boldsymbol{X}^T \boldsymbol{\xi}^t) \ ,$$

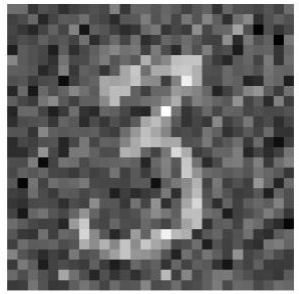
$$C \cong \frac{d}{2\log(d)} \ ,$$

$$C \cong 2^{\frac{d}{2}} \ .$$

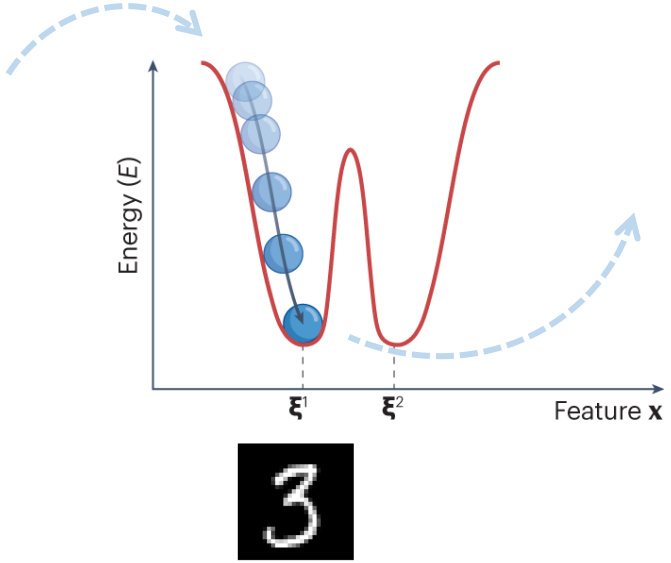
Dmitry Krotov, "A new frontier for Hopfield networks", Nature Reviews Physics, 2023

Latent Modern Hopfield Network

(Modern) Hopfield Networks



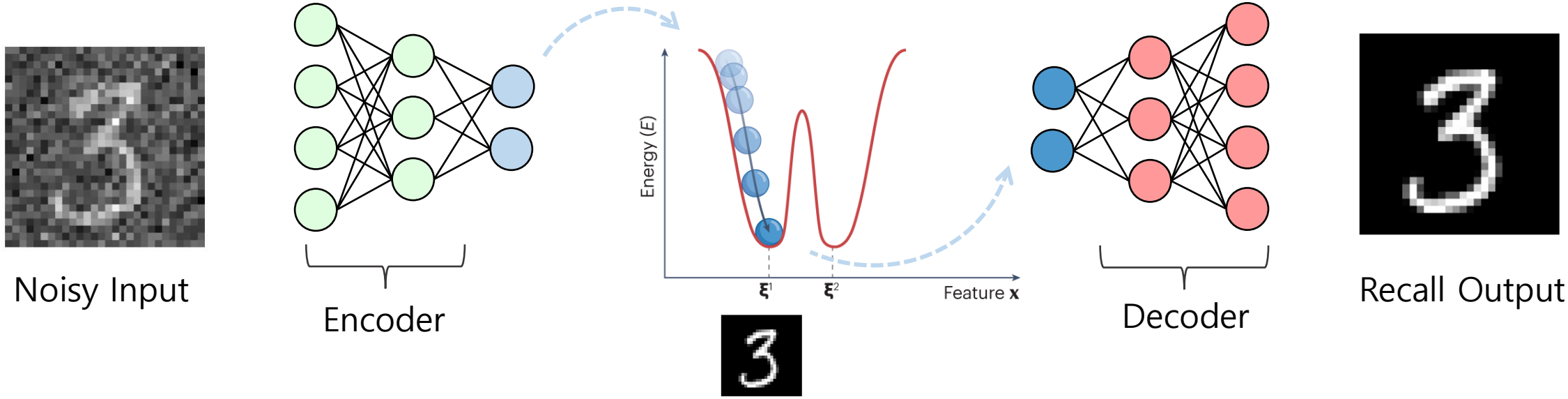
Noisy Input



Recall Output

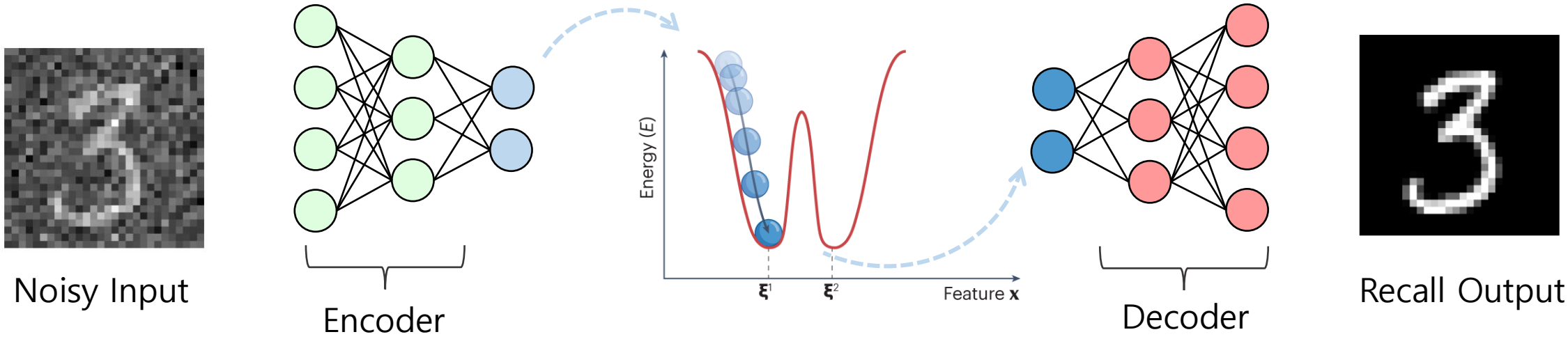
Latent Modern Hopfield Network

Latent Modern Hopfield Networks



Latent Modern Hopfield Network

Latent Modern Hopfield Networks



Biologically	Neocortex	Hippocampus	Neocortex

Intuitively

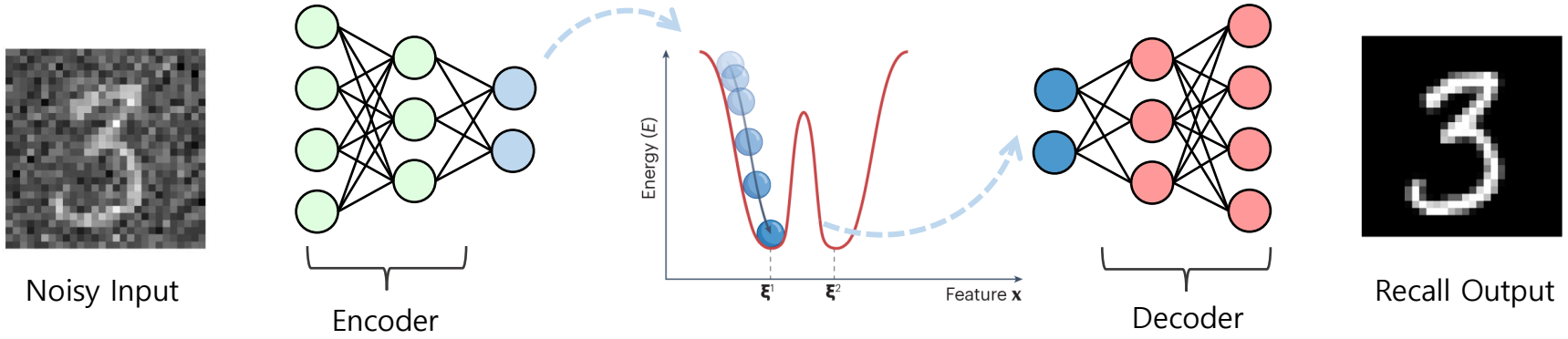
Memory recall in feature space
<- Similar patterns carry similar energy
<- Learn Distance of pattern
<- Higher capacity

Related Work

On the relationship between variational inference and auto-associative memory
- Pretrained encoder & decoder.

Latent structured Hopfield network for semantic association and retrieval
- Recall on classical Hopfield network, use latent alignment loss.

Our model
- Trained end to end
- Recall on modern Hopfield network.



Annabi, L.; Pitti, A.; and Quoy, M. 2022. On the relationship between variational inference and auto-associative memory. Advances in Neural Information Processing Systems, 35: 37497–37509.

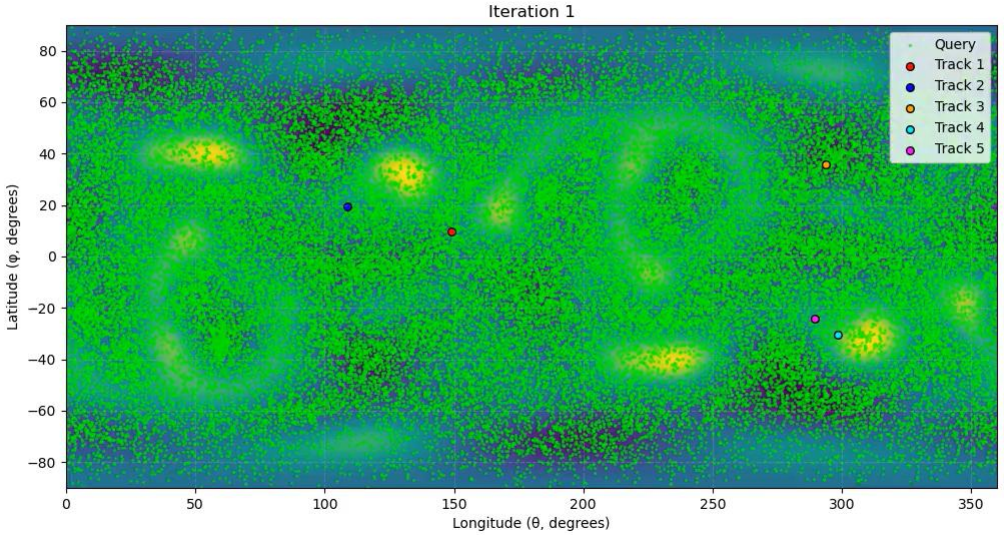
Li, C.; Xue, X.; Feng, J.; and Zeng, T. 2025. Latent Structured Hopfield Network for Semantic Association and Retrieval. arXiv preprint arXiv:2506.01303.

Latent Hopfield Dynamics.

Latent Hopfield Dynamics.

$$z_{t+1} = \sum_{i=1}^{P} \alpha_i(z_t) m_i + \xi_t, \quad \alpha_i(z_t) = \frac{\exp(\beta z_t^\top m_i)}{\sum_{j=1}^{P} \exp(\beta z_t^\top m_j)},$$

$$\begin{aligned} z_t &\in \mathbb{R}^D \\ M &= \{m_1, \ldots, m_P\} \subset \mathbb{R}^D. \\ \xi_t &\sim \mathcal{N}(0, \sigma^2 I) \end{aligned}$$



Training Procedure.

Loss function. $\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \lambda \mathcal{L}_{\text{SW}},$

$$\mathcal{L}_{\text{rec}} = \|x - \hat{x}\|_2^2$$

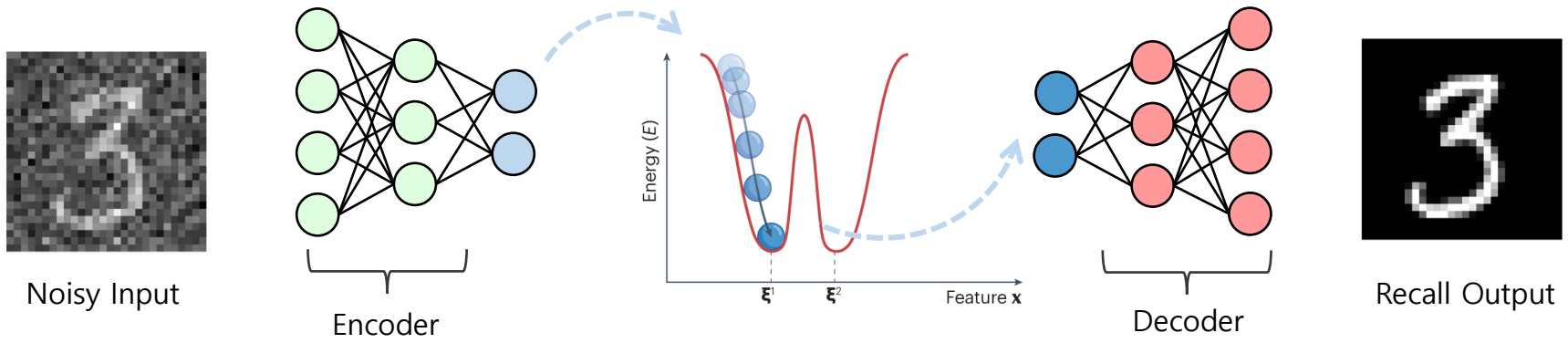
$$\mathcal{L}_{\text{SW}} = \frac{1}{H} \sum_{h=1}^{H} W_2^2(\mathcal{P}_{\boldsymbol{r}_h}(\boldsymbol{\alpha}),\ \mathcal{P}_{\boldsymbol{r}_h}(\mathcal{U})), \qquad \mathcal{P}_{\boldsymbol{r}_h}(\boldsymbol{\alpha}) = \{\langle \boldsymbol{\alpha}_i, \boldsymbol{r}_h \rangle\}_{i=1}^{P}$$

sliced Wasserstein loss

$$\alpha_i(z_t) = \frac{\exp(\beta z_t^\top m_i)}{\sum_{j=1}^{P} \exp(\beta z_t^\top m_j)},$$

$$\boldsymbol{r}_h \sim \mathcal{N}(0, I),$$

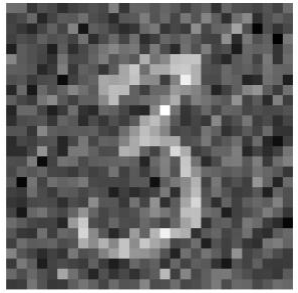
Prevent Latent Collapse !



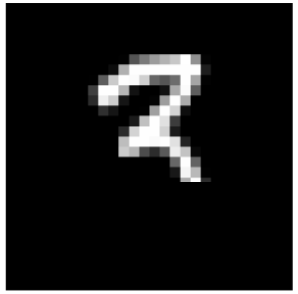
Kolouri, S.; Pope, P. E.; Martin, C. E.; and Rohde, G. K. 2018. Sliced-Wasserstein Autoencoder: An Embarrassingly Simple Generative Model. arXiv:1804.01947

Experiments

Memorization task.



Gaussian Noise



Masked Noise

Target

$$\mathrm{MSE}(x, \hat{x}) < 0.01.$$

Autoencoder
LSHN (latent structured Hopfield network)

MHN (modern Hopfield network)
LMHN (latent modern Hopfield network)

$$n \in \{10, 100, 500, 1000\} \qquad P \in \{64, 1024\}$$

Results

표 1: Pattern recall accuracy (%) under three input conditions: clean (no noise), additive Gaussian noise ($\sigma = 0.3$), and random masking (30% of pixels zeroed out). Each entry is formatted as *No Noise / Gaussian / Mask*. For MHN and LMHN, results are reported for two memory sizes, $P = 64$ and $P = 1024$.

		AE	LSHN	MHN		LMHN (Ours)	
Dataset	$n \backslash P$	-	-	64	1024	64	1024
MNIST	10	**1.000** / 0.700 / 0.000	0.900 / **1.000** / **1.000**	0.700 / 0.700 / 0.700	0.800 / 0.800 / 0.800	**1.000** / 0.967 / 0.933	**1.000** / 0.967 / 0.767
	100	**1.000** / 0.007 / 0.000	0.040 / 0.080 / 0.070	0.290 / 0.280 / 0.290	0.380 / 0.380 / 0.360	0.993 / 0.860 / 0.487	**1.000** / **0.947** / **0.667**
	500	**1.000** / 0.047 / 0.001	0.000 / 0.000 / 0.000	0.064 / 0.064 / 0.062	0.182 / 0.182 / 0.180	0.191 / 0.135 / 0.055	0.987 / **0.877** / **0.987**
	1000	**1.000** / 0.057 / 0.003	0.000 / 0.000 / 0.000	0.055 / 0.054 / 0.052	0.169 / 0.166 / 0.157	0.104 / 0.078 / 0.041	0.832 / **0.653** / **0.318**
CIFAR10	10	**1.000** / 0.120 / 0.440	0.900 / **1.000** / **1.000**	0.867 / 0.867 / 0.800	0.933 / 0.933 / 0.833	**1.000** / 0.967 / 0.967	**1.000** / **1.000** / **1.000**
	100	**1.000** / 0.000 / 0.008	0.550 / 0.080 / 0.410	0.145 / 0.145 / 0.145	0.395 / 0.395 / 0.395	0.270 / 0.193 / 0.177	**1.000** / **0.877** / **0.923**
	500	**1.000** / 0.010 / 0.007	0.422 / 0.000 / **0.150**	0.006 / 0.006 / 0.006	0.082 / **0.082** / 0.080	0.000 / 0.000 / 0.000	0.052 / 0.034 / 0.045
	1000	**0.999** / 0.003 / 0.006	0.020 / 0.000 / 0.013	0.002 / 0.002 / 0.002	0.028 / **0.028** / **0.028**	0.000 / 0.000 / 0.000	0.000 / 0.000 / 0.000

Results

표 1: Pattern recall accuracy (%) under three input conditions: clean (no noise), additive Gaussian noise ($\sigma = 0.3$), and random masking (30% of pixels zeroed out). Each entry is formatted as *No Noise / Gaussian / Mask*. For MHN and LMHN, results are reported for two memory sizes, $P = 64$ and $P = 1024$.

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	100	**1.000** / 0.000 / 0.008	0.550 / 0.080 / 0.410	0.145 / 0.145 / 0.145	0.395 / 0.395 / 0.395	0.270 / 0.193 / 0.177	**1.000** / **0.877** / **0.923**
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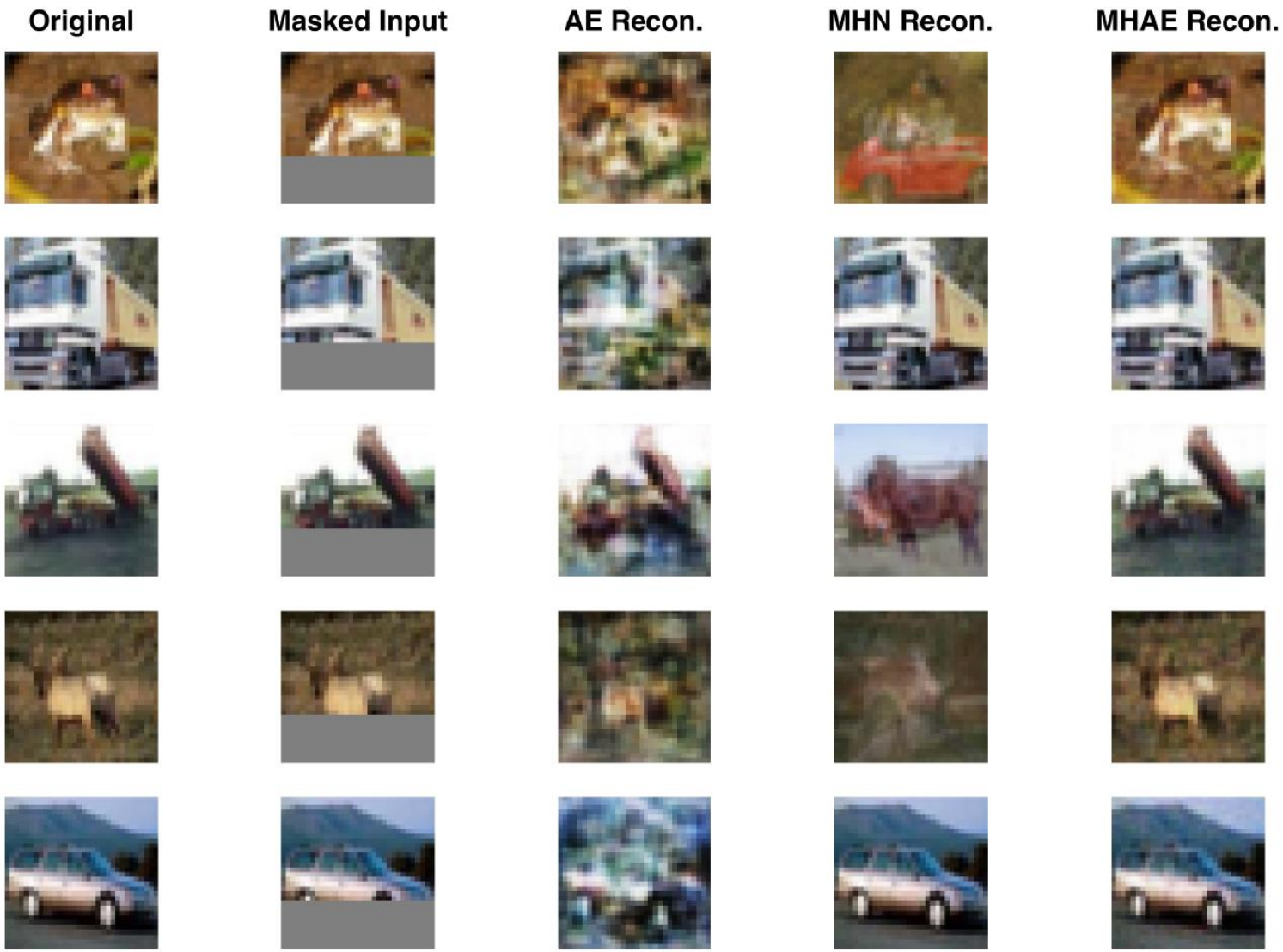
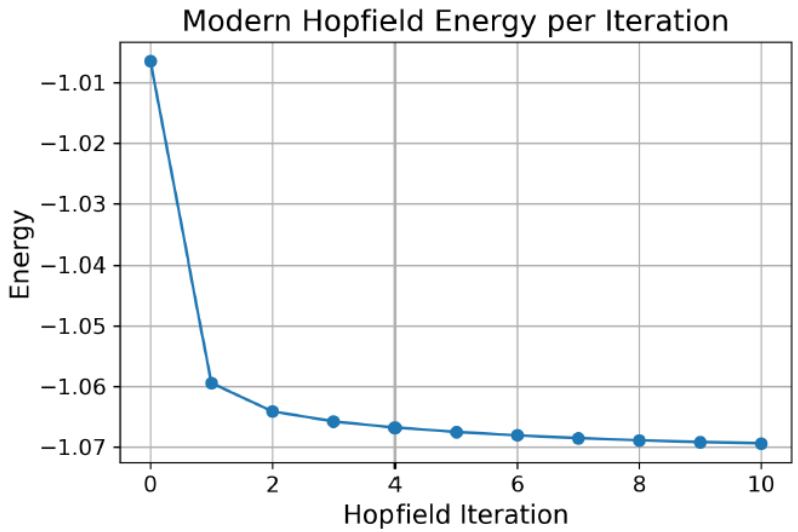
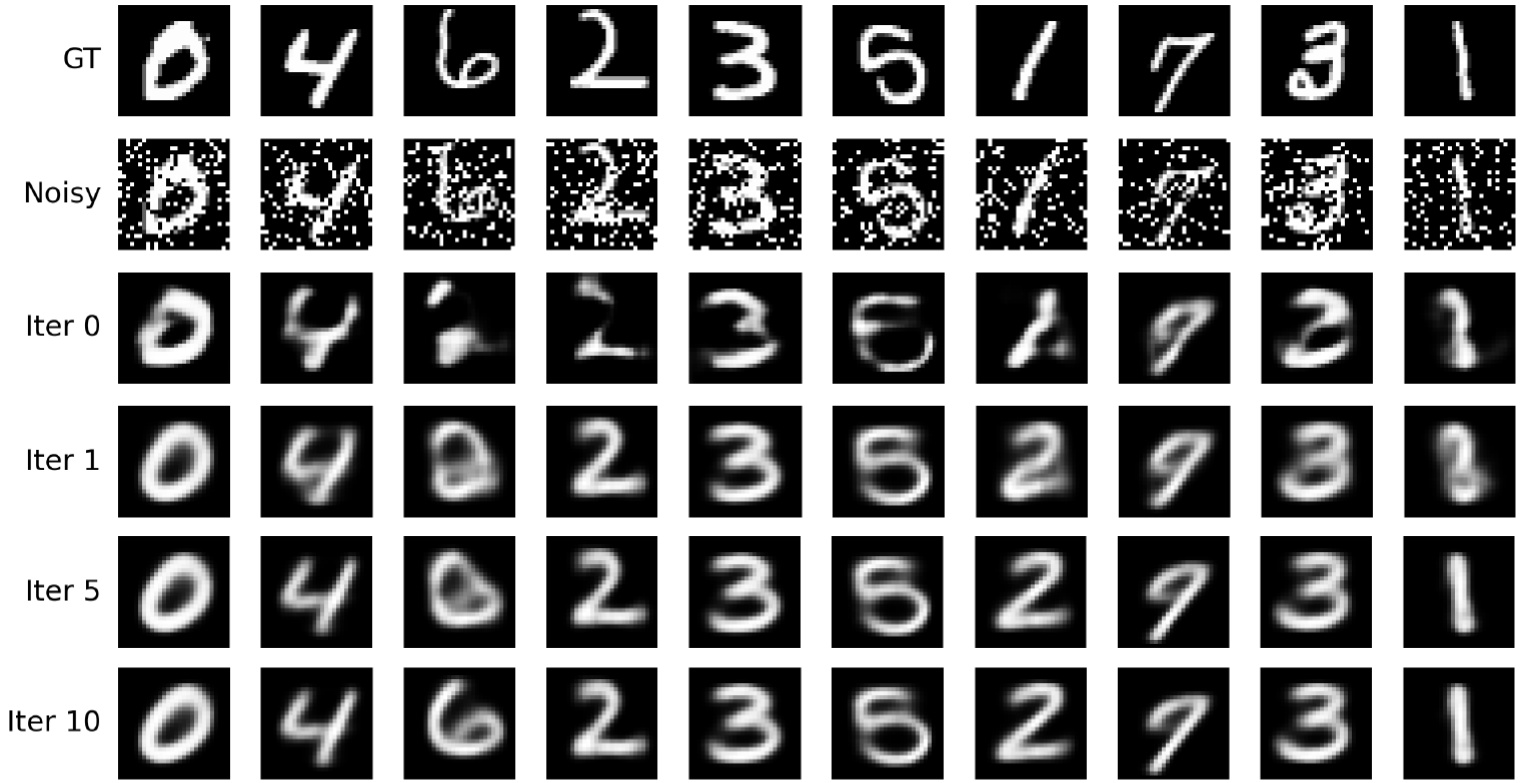
Results



Results


Modern Hopfield Energy per Iteration
Energy
−1.01
−1.02
−1.03
−1.04
−1.05
−1.06
−1.07
0
2
4
6
8
10
Hopfield Iteration


GT
Noisy
Iter 0
Iter 1
Iter 5
Iter 10

Conclusion

- In this work, we proposed the Latent Modern Hopfield Network (LMHN), which integrates Modern Hopfield Network dynamics into the latent space, trained end to end.

- LMHN captures abstract and generalizable features while preserving strong associative recall ability.